ALUMINUM ALLOY CONDUCTORS (TYPE- AL59) SS 424 08 14

Nominal Area	Sectional Area	Equivalent copper area	Stranding		Diameter of Complete Conductor	Weight	Rated Strength	DC Resistance @ 20°C	Current Capacity	
			No. of Aluminium Wires	Individual wire diameter					@ 75°C	@ 85°C
(mm²)	(mm²)	(mm²)	(No.)	(mm)	(mm)	(Kg/Km)	KN	(Ω/Km)	(Ampere)	(Ampere)
31	31.14	18.48	7	2.38	7.14	85	7.77	0.943	120	142
62	62.44	37.06	7	3.37	10.11	170	15.6	0.47	181	218
99	99.3	58.93	7	4.25	12.80	271	22.8	0.296	238	289
157	158.59	94.12	19	3.26	16.3	436	39.7	0.186	313	384
241	241.16	143.13	19	4.02	20.1	663	55.5	0.123	397	493
329	330.03	195.87	37	3.37	23.6	910	82.5	0.0899	475	595
454	454.49	269.74	61	3.08	27.72	1260	113	0.0654	567	719
593	593.62	352.31	61	3.52	31.68	1640	143	0.0501	655	840
774	774.24	459.51	61	4.02	36.18	2140	178	0.0384	754	977
910	910.74	540.52	61	4.36	39.2	2520	209	0.0326	819	1070

ECO CONDUCTORS

Eco Series conductors consists of trapezoidal shaped special Aluminium Alloy (AL-59) containing Magnesium, Silicon and Copper stranded with round core. The trapezoidal shape provides up to 96% compactness.

REF. SPECIFICATION - SS 42408013, 14

Code Word	Size	Appropriate System Voltage	Equivalent ACSR	Reference Standards	Conductor diameter	Weight	Ultimate tensile strength	DC resistance at 20°C temperature	Current carrying capacity at maximum operating temperature	Power Transfer Capacity (at 33 kV at 0.95 PF)
	Sq. mm.	kV			mm	kg/km	kN	Ohm/km	A	MW/CKT
ECO DOG	139	S110 kV	DOG	SS 42401813, SS 42401814 & IEC 62219 & IS 398 Part IV	14.15	380	31.0	0.2181	393	21
ECO PANTHER	310	S132 kV	PANTHER		21.2	855	64.8	0.0987	641	34
ECO PANTHER	353	S132 kV	PANTHER		22	973	81.95	0.08	698	37